

WILBARGER COUNTY AP, TX, USA

WMO: 720281

Lat: 34.226N

Lon: 99.284W

Elev: 386

StdP: 96.78

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53969

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.5	-5.0	-17.3	0.9	-0.2	-14.6	1.1	0.9	13.3	8.5	11.9	9.7	3.6	20	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	40.9	21.5	39.0	21.6	37.7	21.5	24.6	32.9	24.0	32.3	23.5	31.9	5.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	18.2	26.9	22.2	17.7	26.7	21.3	16.7	26.7	76.3	32.6	73.8	32.1	71.8	31.3	35.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.8	10.6	9.2	-11.0	42.1	2.9	2.0	-13.1	43.5	-14.8	44.7	-16.4	45.8	-18.5	47.3
			-12.0	25.1	2.7	1.0	-13.9	25.9	-15.5	26.4	-17.1	27.0	-19.1	27.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.3	6.1	8.2	13.5	17.5	22.9	28.2	29.8	29.5	25.6	18.7	12.5	6.6	
	DBStd	9.70	5.14	5.95	5.60	4.91	4.68	2.95	2.74	3.07	3.53	4.98	5.01	5.10	
	HDD10.0	431	141	94	29	5	0	0	0	0	0	4	31	127	
	HDD18.3	1523	379	286	168	72	15	0	0	0	1	58	181	364	
	CDD10.0	3466	20	44	137	230	401	547	614	603	468	272	107	22	
	CDD18.3	1517	0	3	17	47	158	297	356	345	219	68	7	1	
	CDH23.3	19707	14	59	231	624	1848	3872	5002	4687	2505	741	111	12	
	CDH26.7	10977	2	14	65	245	917	2239	3053	2837	1294	284	24	2	
Wind	WSAvg	4.3	4.2	4.5	5.0	5.2	4.7	4.8	4.0	3.7	3.7	4.1	4.3	4.0	
Precipitation	PrecAvg	693	29	35	49	66	86	102	61	62	63	65	40	31	
	PrecMax	1077	63	103	97	162	193	272	195	138	162	215	182	115	
	PrecMin	413	1	0	0	3	14	3	4	0	8	1	1	0	
	PrecStd	182	21	29	30	45	48	72	49	40	46	53	45	32	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.3	28.5	31.7	35.6	38.8	42.2	42.4	42.8	38.9	34.1	29.9	25.0	
		MCWB	11.2	14.1	15.5	17.9	18.8	21.1	21.5	21.5	20.8	19.6	16.4	13.5	
	2%	DB	21.4	24.8	28.2	32.3	36.6	39.0	40.2	40.9	36.9	32.1	26.4	21.0	
		MCWB	10.3	13.1	15.7	17.7	20.1	21.8	21.7	21.8	20.8	19.4	14.9	11.0	
	5%	DB	18.2	21.4	26.1	29.5	34.2	37.3	38.8	39.1	35.1	30.0	23.9	17.7	
		MCWB	9.8	11.7	14.9	17.0	20.2	21.6	21.7	21.9	21.0	18.2	14.3	10.3	
	10%	DB	15.5	18.1	23.1	27.1	32.0	35.9	37.4	37.4	33.2	27.6	21.2	15.2	
		MCWB	7.9	10.9	14.0	16.5	19.7	21.9	21.7	22.0	20.6	17.3	13.5	9.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	16.8	19.7	23.0	24.1	25.3	26.2	25.1	24.5	22.8	18.8	16.3
			MCDB	20.0	21.8	25.3	30.1	31.4	34.1	34.2	34.2	31.8	29.0	25.5	20.6
2%		WB	12.2	15.1	18.5	20.5	23.1	24.2	24.5	24.2	23.6	21.6	17.2	13.8	
		MCDB	17.8	21.2	24.0	26.8	31.3	33.0	32.3	33.1	30.2	28.0	22.4	18.1	
5%		WB	10.2	13.1	17.3	19.4	22.4	23.6	23.8	23.7	22.9	20.5	16.0	11.4	
		MCDB	17.0	20.1	22.5	25.7	30.1	32.7	31.8	32.7	29.2	26.6	21.6	16.7	
10%		WB	8.4	11.1	15.6	18.1	21.5	23.0	23.1	23.1	22.3	19.3	14.6	9.3	
		MCDB	15.4	18.3	21.2	24.2	28.8	32.2	31.4	32.3	29.0	25.2	20.0	15.2	
Mean Daily Temperature Range	5% DB	MDBR	13.7	13.5	14.3	14.5	13.4	12.9	13.1	13.1	12.9	13.7	13.7	12.5	
		MCDBR	19.2	19.4	17.9	18.4	16.5	15.1	15.3	15.3	14.9	16.2	17.3	17.3	
		MCWBR	10.4	9.7	8.4	8.2	5.4	4.2	3.5	3.7	3.9	6.2	8.5	9.8	
	5% WB	MCDBR	17.4	16.5	13.3	14.7	13.5	13.0	12.2	12.9	11.4	12.2	14.2	15.1	
		MCWBR	10.4	9.8	7.2	8.1	5.5	4.5	3.8	3.3	3.5	5.9	8.2	9.8	
Clear-Sky Solar Irradiance	taub	0.286	0.299	0.328	0.364	0.390	0.418	0.417	0.410	0.379	0.348	0.312	0.288		
	taud	2.535	2.506	2.442	2.359	2.358	2.315	2.365	2.328	2.391	2.462	2.495	2.529		
	Ebn at Noon	935	956	948	921	896	867	865	867	881	885	891	910		
	Edh at Noon	81	93	107	122	124	130	123	125	112	96	84	77		
All-Sky Solar Radiation	RadAvg	3.07	3.67	4.90	5.94	6.38	7.02	6.99	6.38	5.31	4.12	3.20	2.62		
	RadStd	0.31	0.50	0.31	0.46	0.66	0.58	0.41	0.38	0.37	0.56	0.40	0.27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page